DIFFERENTIATION OF COMMON FUNCTIONS

A general rule of differentiating $y = ax^n$ can be given, where a and n are constants.

The rule is: If $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$

or If $f(x) = ax^n$ then $f'(x) = anx^{n-1}$.

This is true for all real values of a and n.

For example, If $y = 4x^3$ then $a = 4$ and $n = 3$, and

$$\frac{dy}{dx} = anx^{n-1} = (4)(3)x^{3-1} = 12x$$

If $y = ax^n$ and $n = 0$ then $y = ax^0$

$$\frac{dy}{dx} = (a)(0)x^{0-1} = 0.$$

STANDARD DERIVATIVES

The Standard derivatives are summarized below although they may be proved theoretically and are true for all real values of x

y or $f(x)$	dy/dx or $f'(x)$
ax^n	anx^{n-1}
$\sin ax$	$a\cos ax$
$\cos ax$	$-a\sin ax$
e^{ax}	ae^{ax}
$\ln ax$	$\frac{1}{x}$

The differential coefficient of a sum or difference is the sum or difference of the differential coefficient of the separ terms.

If $f(x) = p(x) + q(x) - r(x)$. Where f, p, q and r are f

Then $f'(x) = p'(x) + q'(x) - r'(x)$

Example 1

Find the differential coefficient of

$y = 12x^3$.

Since $y = 12x^3$ then

$$\frac{dy}{dx} = (12)(3)x^{3-1} = 36x^2$$

Example 2

Find the differential coefficient of $y = \frac{12}{x^3}$

$$y = \frac{12}{x^3} = 12x^{-3}$$

$$\therefore \frac{dy}{dx} = (12)(-3)x^{-3-1} = 36x^{-4} = -\frac{36}{x^4}$$

Example 3

differentiate $y = 6x$

$$\frac{dy}{dx} = 6(1)x^{1-1} = 6x^0 = 6$$

EXAMPLE

Problem 4: Find the derivatives of

(a) $y = 3\sqrt{x}$ (b) $y = \dfrac{5}{\sqrt[3]{x^4}}$

$y = 3\sqrt{x}$ which we can rewrite in standard form as

$$y = 3x^{1/2}$$

$$\therefore \frac{dy}{dx} = (3)(\tfrac{1}{2})x^{\frac{1}{2}-1} = \frac{3}{2}x^{-\frac{1}{2}}$$

$$\text{Thus } \frac{dy}{dx} = \frac{3}{2x^{1/2}} = \frac{3}{2\sqrt{x}}$$

(b) $$y = \frac{5}{\sqrt[3]{x^4}} = \frac{5}{x^{4/3}} = 5x^{-4/3}$$

$$\text{Thus } \frac{dy}{dx} = 5\left(-\frac{4}{3}\right)x^{-\frac{4}{3}-1} = \frac{-20}{3}x^{-7/3}$$

$$= \frac{-20}{3x^{7/3}} = -\frac{20}{3\sqrt[3]{x^7}}.$$

Problem 5: Differentiate, wrt x

$$y = 5x^4 + 4x - \frac{1}{2x^2} + \frac{1}{\sqrt{x}} - 3$$

We rewrite:

$$y = 5x^4 + 4x + \tfrac{1}{2}x^{-2} + x^{-\frac{1}{2}} - 3$$

When differentiating we consider each term

$$\frac{dy}{dx} = (5)(4)x^{4-1} + (4)(1)x^{1-1} - \tfrac{1}{2}(-2)x^{-2-1} + (1)(-\tfrac{1}{2})x^{-\frac{1}{2}-1} - 0$$

$$= 20x^3 + 4 + x^{-3} - \tfrac{1}{2}x^{-3/2}$$

i.e $\frac{dy}{dx} = 20x^3 + 4 + \frac{1}{x^3} - \frac{1}{2\sqrt{x^3}}$

Example 6: Find the differential coefficients of
(a) $y = 3\sin 4x$ (b) $f(t) = 2\cos 3t$

When $y = 3\sin 4x$

$\frac{dy}{dx} = (3)4\cos 4x$

$= 12\cos 4x$

When $f(t) = 2\cos 3t = y.$

$f'(t) = \frac{dy}{dt} = (2)(-3\sin 3t)$

$= -6\sin 3t$

Example 7. Determine the derivatives of
(a) $y = 3e^{5x}$ (b) $f(\theta) = \frac{2}{e^{3\theta}}$ (c) $y = 6\ln 2x$

(a)
When $y = 3e^{5x}$

$\frac{dy}{dx} = (3)(5)e^{5x} = 15e^{5x}$

(b) $f(\theta) = \frac{2}{e^{3\theta}} = 2e^{-3\theta}$

$\therefore f'(\theta) = (2)(-3)e^{-3\theta} = -6e^{-3\theta} = \frac{-6}{e^{3\theta}}$

d the grad
y
(1,4)

nd the gradient of a curve ~~at a given pt~~

$y = 3x^4 - 3x^2 + 5x - 2$. at the points $(0,-2)$ an

$(1,4)$

Since $y = 3x^4 - 2x^2 + 5x - 2$

$\therefore$ the gradient $= \frac{dy}{dx} = 12x^3 - 4x + 5$

At the point $(0,-2)$ $x = 0$

$\therefore$ $12(0)^3 - 4(0) + 5 = 5$

At the point $(1,4)$, $x = 1$

thus the gradient $= 12(1)^3 - 4(1) + 5 = 13$.

Example 9. Determine the co-ordinates of the point on the graph $y = 3x^2 - 7x + 2$ where the gradient is -1

When $y = 3x^2 - 7x + 2$ then $\frac{dy}{dx} = 6x - 7$

Since the gradient is -1 then $6x - 7 = -1$

from which $x = 1$

When $x = 1$, $y = 3(1)^2 - 7(1) + 2 = -2$

EXAMPLE 10 SOLVE $y = \frac{e^x - e^{-x}}{2}$

$y = \frac{e^x}{2} - \frac{e^{-x}}{2} = \frac{1}{2}e^x - \frac{1}{2}e^{-x}$

PLE 11

$\therefore \frac{dy}{dx} = (\frac{1}{2})(1)e^{x} - \frac{1}{2}(-1)e^{-x}$

$\frac{dy}{dx} = \frac{1}{2}e^{x} + \frac{1}{2}e^{-x}$

$\therefore \frac{dy}{dx} = \frac{e^{x} + e^{-x}}{2}$

DIFFERENTIATION OF A PRODUCT

When $y = uv$ and u and v are functions of x.

then $\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{dy}{dx}$

Example 10 find the differential coefficient of $y = 3x^2 \sin 2x$.

$3x^2 \sin 2x$ is the product of two terms $3x^2$ and $\sin 2x$

$\therefore u = 3x^2$ and $v = \sin 2x$.

Using the product rule:

$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$

$= \frac{dy}{dx} = (3x^2)(2\cos 2x) + (\sin 2x)(6x)$

$\frac{dy}{dx} = 6x^2\cos 2x + 6x\sin 2x$

$= 6x(x\cos 2x + \sin 2x)$

mple 11: Find the rate of change of y w.r.t x (7)

ven $y = 3\sqrt{x}\ln 2x$

The rate of change of y with respect to x is given by dy/dx

$y = 3\sqrt{x}\ln 2x = 3x^{1/2}\ln 2x$ This is a product rule.

$u = 3x^{1/2}$ and $v = \ln 2x$

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$= (3x^{1/2})(1/x) + \ln 2x\left[3(1/2)x^{1/2-1}\right]$$

$$= 3x^{1/2-1} + \ln 2x\left(3/2\right)x^{-1/2}$$

$$\cancel{= 3x^{-1/2}(1+}$$

$$= 3x^{-1/2} + \ln 2x\left(3/2\right)x^{-1/2}$$

$$= 3x^{-1/2}\left(1 + 1/2\ln 2x\right).$$

$$\frac{dy}{dx} = \frac{3}{\sqrt{x}}\left(1 + 1/2\ln 2x\right).$$

EXAMPLE 12 Differentiate $y = x^3\cos 3x\ln x$

let $u = x^3\cos 3x$ and $v = \ln x$

then $$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$\frac{du}{dx} = x^3(-3\sin 3x) + (\cos 3x)(3x^2)$$

$$\frac{dv}{d} = \frac{1}{x}$$

$$\therefore \frac{dy}{dx} = x^3 \cos 3x \left(\frac{1}{x}\right) + \ln x \left[-3x^3 \sin 3x + 3x^2 \cos 3x\right]$$

$$= x^2 \cos 3x + 3x^2 \ln x \left[\cos 3x - x \sin 3x\right].$$

$$\frac{dy}{dx} = x^2 \{\cos 3x + 3 \ln x (\cos 3x - x \sin 3x)\}.$$

DIFFERENTIATION OF A QUOTIENT

When $y = \frac{u}{v}$ and both u and v are functions of x.

then $$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

This is called the quotient rule.

Example 14. Find the differential coefficient of

$$y = \frac{4 \sin 5x}{5x^4}$$

$y = \frac{4 \sin 5x}{5x^4}$ let $u = 4 \sin 5x$ and $v = 5x^4$

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

$$\frac{du}{dx} = 20 \cos 5x \text{ or } (4)(5) \cos 5x = 20 \cos 5x$$

$$\frac{dv}{dx} = (5)(4)x^3 = 20x^3$$

$$\therefore \frac{dy}{dx} = \frac{5(x^4)(20 \cos 5x) - (4 \sin 5x)(20x^3)}{(5x^4)^2}.$$

$$\frac{dy}{dx} = \frac{100x^4 \cos 5x - 80x^3 \sin 5x}{25x^8}$$

EXAMPLE 17 Differentiate $y = \frac{te^{2t}}{2\cos t}$ Dont teach

Let $u = te^{2t}$ and $v = 2\cos t$

$$\frac{du}{dt} = (t)2e^{2t} + (e^{2t})(1)$$

$$\frac{dv}{dt} = -2\sin t$$

Hence $\frac{dy}{dt} = \frac{v\frac{du}{dt} - u\frac{dv}{dt}}{v^2}$.

$$= \frac{(2\cos t)(2te^{2t} + e^{2t}) - (te^{2t})(-2\sin t)}{(2\cos t)^2}$$

$$= \frac{4te^{2t}\cos t + 2e^{2t}\cos t + 2te^{2t}\sin t}{4\cos^2 t}$$

$$= \frac{2e^{2t}[2t\cos t + \cos t + t\sin t]}{4\cos^2 t}$$

i.e $\frac{dy}{dx} = \frac{e^{2t}}{2\cos^2 t}(2t\cos t + \cos t + t\sin t)$

FUNCTION OF A FUNCTION

If y is a function of x then $\frac{dy}{dx} = \frac{dy}{du} * \frac{du}{dx}$

This is called the function of function rule or the Chain rule.

EXAMPLE 17

$$\frac{dy}{dx} = 36 \tan^3 3x \sec^2 3x$$

EXAMPLE 20 find the differential coefficient of

$$y = \frac{2}{(2t^2-5)^4}$$